Observer-Based Variable Impedance Control Using Moving Horizon Estimation for Robot Machining Thin-Walled Workpieces

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***Abstract*—Adaptability is one of the most important survival and living abilities in nature and human society, which needs to perceive the environment and regulate self-behaviors. To empower robots in manufacturing with this ability, an observer-based variable impedance control scheme is proposed in this article. It senses changes in environmental properties during contact and utilizes the observations to guide the decisions of the robot controller. Specifically, we developed a real-time moving horizon based robot-environment observer, which uses historical information to predict the current impedance parameters of the external contact environment and robot internal joint disturbances simultaneously. Then, a variable impedance control law is designed to eliminate state tracking errors and decrease machining vibration. The stability of the closed-loop system is theoretically proven. Experiments on machining thin-walled workpieces validate the efficacy of the proposed method, where the robot exhibits compliant force behavior resulting in superior workpiece surface roughness among various comparison methods.**

***Index Terms*—Intelligent and flexible manufacturing, moving horizon estimation, robot-environment interaction, robotic machining, variable impedance control.**

I. Introduction

THIN-WALLED workpieces [1] are widely used flexible elements for mechanical structures, such as aircraft skin [2], siding of hulls [3], and connection plates of trains [4]. As depicted in Fig. 1, such workpieces commonly have a long size but slim profile, leading to low stiffness, and inadequate clamping. Robotic machining [5] of thin-walled workpieces can reduce the burden of human labor in industry [6]. The advantages of robots lie in their flexible operation, feasible motion accuracy, and convenient sensor integration, which are beneficial for yielding high machining quality. However, the rigidity of the robot is relatively low compared to the machine tools, and the material removal processes commonly encounter dramatic interacting forces between the robot and the environment resulting in workpiece deformation, vibrations, and even unstable chatters [7]. The intrinsic instability during machining involves unknown environmental properties, e.g., workpiece impedance, robot motor disturbances, and so on. There are many factors that cannot be directly seen and measured that will worsen the robotic machining performance.

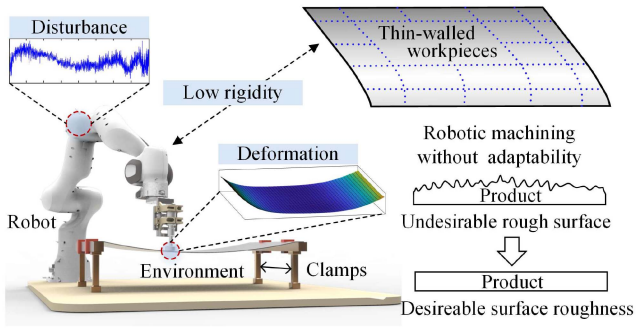


Fig. 1. Robotic machining of thin-walled workpieces. Robot disturbances and workpiece deformation appeared in the machining due to the low rigidity of the robot and workpiece, resulting in an undesirable rough surface of the product.

Offline and online control methods have been studied to improve the machining quality of those kinds of workpieces. Klimchik et al. [8] proposed a classical offline scheme, which utilized an offline prediction model to estimate the machining force through the desired tool trajectory and then applied a compliance model to modify the nominal tool trajectory in the presence of the machining forces [9]. In addition, force-induced deformations can also be predicted using finite element analysis technology before machining [10]. Offline methods require accurate knowledge of workpiece material and surface location for calculation of the possible workpiece deformation. Stochastic factors, such as machining disturbance or uneven material properties, are hardly considered in offline strategies. Then, online control strategies are introduced for compensating the

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deformation-caused errors in real-time with stochastic factors. Based on the directly measured information, Yuan et al. [1] introduced a data-driven method for workpiece deformation prediction. Bi et al. [11] developed a fuzzy support vector machine-based method to decompose the machining errors. However, robotic machining systems are different from machine tool systems. They have lower control response and rigidity. It is more inclined to use force feedback technology to ensure the machining accuracy of the robot. Impedance control [12] is then considered and widely used in robot machining tasks [13]. It adjusts the relationship between the force response and environment deformation through the principle of a mass-spring-damper model. Nevertheless, robot controllers with fixed impedance parameters are conservative. The instability may occur when the environmental impedance changes. Therefore, the robot's adaptability for machining thin-walled workpieces is still limited.

Based on the abovementioned analysis, a more environment-adaptable controller is desirable for the thin-walled workpiece machining task. Recently, Li et al. [14] investigated that humans could skillfully interact with the environment for different purposes by adapting their impedance, which inspired us to consider variable impedance control (VIC). Ikeura and Inooka [15] first introduced a variable damping scheme for a robot to work with a human in cooperative lifting tasks, which was later improved by Rahman et al. [16] to include variable stiffness. In manipulation tasks, VIC strategies are commonly designed based on trajectory tracking errors [17]. Recently, learning methods such as imitation learning [18], [19] iterative learning [20], and reinforcement learning [21], [22] have been investigated, accounting for adaptive impedance for tasks, especially with humans in the loop [23]. The stability of VIC has been studied from the passivity concept [24] and extended in [25]. The state-independent stability conditions for VIC are studied in [26]. The variable impedance profiles are generally restricted by the stability of the closed-loop system [27]. In contrast, Papastavridis and Dai [28], [29], and Papastavridis et al. [30] presented stable VIC frameworks that enable unconstrained impedance gains for flexible-joint robots. The aspect of robot redundancy is often neglected in VIC, but it was taken into account in [31] to design a passive controller. Considering VIC in robot machining can improve robot environmental adaptability, resulting in better surface roughness [1] and minor force fluctuations [32] than fixed impedance strategies. In fact, the performance of the force-induced interaction depends on the impedance relationship between the robot and the environment. Thus, we aim to study a variable impedance strategy in which robot impedance varies with the workpiece impedance during interaction to improve the machining quality. A prerequisite of this goal is to obtain the actual environment impedance in real-time, which motivates us to consider estimation methods.

Among a number of estimation methods [33], moving horizon estimation (MHE) is an optimization-based method [34], which utilizes a window of historical measurements to predict the current unmeasurable states and model parameters [35]. A real-time MHE algorithm for simultaneously estimating states and parameters was studied in [36]. Kraus et al. [37] used constrained MHE to estimate traction and slipping parameters for an autonomous vehicle. Liu et al. [38] proposed a multirate MHE for mobile robots to upgrade the estimation performance when some sensor measurements are missing. By combining MHE and dynamic mode decomposition methods, the flow-regime transitions can be detected [39]. Julian et al. [40] introduced the property of sub-optimal solution in model predictive control (MPC) to ensure the stability of the MHE. Constraints always exist for real robots due to physical limitations and safety concerns. The motivation for choosing MHE in this article lies in its multiobjective properties and convenience in setting constraints. Hence, both the external environment impedance parameters and internal disturbances for the robot machining process can be acquired simultaneously in one constrained optimization problem.

In this article, an observer-based variable impedance control method for thin-walled workpiece machining tasks is proposed. The variable impedance of the external environment, as well as the internal model disturbances, are simultaneously estimated by a customized MHE algorithm. Then, the observations are utilized to establish a robot variable impedance control law with model disturbances compensation. The robot's impedance varies with the change in environmental impedance, which improves the adaptability of the robot for interacting with the impedance-changing environment. The specific contributions offered by this article are presented as follows.

1) The scheme of moving horizon estimation-based variable impedance control (MHE-VIC) is proposed to improve robot adaptability in force interaction tasks, especially for improving the machining quality of the thin-walled workpiece, where the robot impedance can be regulated in accordance with the estimated environmental impedance.
2) A novel robot-environment observer is developed by using MHE, which can simultaneously estimate the impedance of the external contact environment and the internal robot joint disturbances. Multiunknown parameters estimation and explicit physical constraints are the advantages of the observer. It utilizes robot historical information to estimate the current unknown parameter accurately.
3) The variable impedance controller is designed based on the observations to adapt to changes in the impedance of the contact environment and avoid joint disturbances during interaction. It ensures the convergence of the state tracking errors and decreases the vibration in the machining tasks. The stability analysis of the closed-loop system using the Lyapunov theory is given.
4) The practical implementation of the whole control scheme is constructed on a seven-DOF robot within a 1 ms control cycle. Comprehensive and comparative results of thin-walled workpiece machining experiments for different controllers demonstrate that both the control accuracy and machining quality are improved.

The rest of this article is organized as follows. Section II illustrates the idea of the proposed MHE-VIC strategy for making the robot more adaptable to the environment. Section III comprises the formulation of the robot-environment observer and variable impedance controller with analysis and stability

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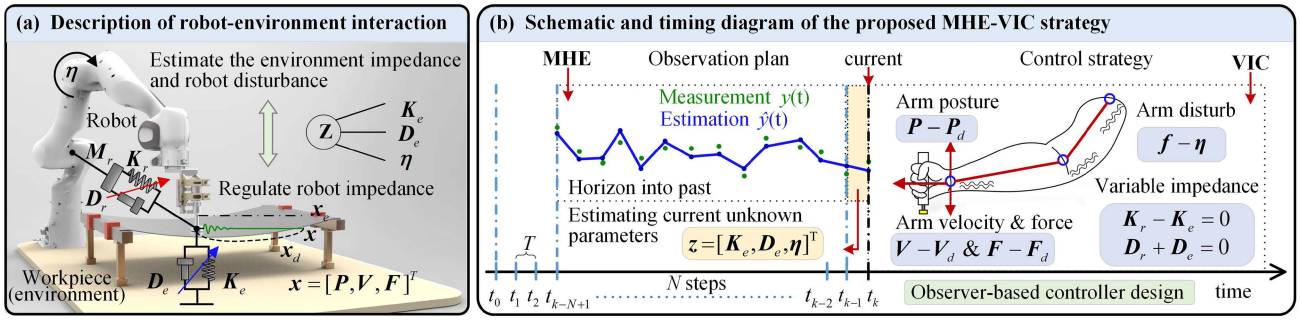


Fig. 2. (a) Description of the robot-environment interaction with impedance strategy. (b) Schematic and timing diagram of the proposed MHE-VIC strategy.

proof. Section IV validates and discusses the practical effect of the proposed method in grinding tasks compared with different controllers. Finally, Section V concludes this article.

II. Interpretation of Robot-Environment Interaction Strategy

The representative scenario of robotic machining of the thin-walled workpiece is shown in Fig. 2(a). As industry demands have evolved, workpieces have become large and thinner, leading to incomplete clamping and low rigidity. The state matrix of the robot is established as $\boldsymbol{x} = [\boldsymbol{P}, \boldsymbol{V}, \boldsymbol{F}]^{\mathrm{T}}$, where $\boldsymbol{P}, \boldsymbol{V}$, and $\boldsymbol{F} \in \mathbb{R}^m$ denote the actual position, velocity, and external force of the robot end-effector in Cartesian space, respectively. m is the dimension of the Cartesian space. The objectives for machining, such as grinding, those components can be described as maintaining the desired position, tangential feed velocity, and normal grinding force. Thus, the state tracking error at the time t_k in the robot machining task can be defined as

$$\boldsymbol{e}(t_k) = \boldsymbol{x}(t_k) - \boldsymbol{x}_d(t_k). \tag{1}$$

From a general perspective, impedance control can provide the robot with compliant force behavior for consistent machining quality. However, the classical impedance model of using robot position to regulate contact force is not suitable for accurate tracking tasks. In order to eliminate the tracking errors in (1), the force tracking Cartesian impedance model [41], [42] is selected in this article, which is described as

$$\boldsymbol{M}_r(\ddot{\boldsymbol{P}} - \ddot{\boldsymbol{P}}_d) + \boldsymbol{D}_r(\dot{\boldsymbol{P}} - \dot{\boldsymbol{P}}_d) + \boldsymbol{K}_r(\boldsymbol{P} - \boldsymbol{P}_d) = \boldsymbol{F} - \boldsymbol{F}_d \tag{2}$$

where $\boldsymbol{M}_r, \boldsymbol{D}_r$, and $\boldsymbol{K}_r \in \mathbb{R}^{m\times m}$ are the mass, damping, and stiffness matrices of the robot in Cartesian space, respectively, also named impedance parameters. $\boldsymbol{P}_d$ and $\boldsymbol{F}_d \in \mathbb{R}^m$ denote the desired position and external force of the robot end-effector.

As for the contact environment, the impedance model is generally established as a spring and damper. The reaction force caused by the deformation of the environment, which is equivalent to the external force of the robot, can be expressed as

$$\boldsymbol{F} = \boldsymbol{K}_e(\boldsymbol{P}_e - \boldsymbol{P}) + \boldsymbol{D}_e(\dot{\boldsymbol{P}}_e - \dot{\boldsymbol{P}}) \tag{3}$$

where $\boldsymbol{K}_e$ and $\boldsymbol{D}_e \in \mathbb{R}^{m\times m}$ denote the stiffness and damping matrices of the environment, respectively. The vector $\boldsymbol{P}_e \in \mathbb{R}^m$ specifies the surface location of the contact environment in the normal direction, as well as the desired time-varying trajectory of the robot in other directions.

By analyzing (2), the actual position of the robot end-effector $\boldsymbol{P}$ should converge to the desired value $\boldsymbol{P}_d$ to make the robot's external force $\boldsymbol{F}$ converge to the desired value $\boldsymbol{F}_d$. To produce a constant contact force at a specific position in the workpiece nominal direction, the stabilized desired position of the robot end-effector is specified to deform the environment by a constant amount. Thus, we assume that the components in $\ddot{\boldsymbol{P}}_d$ and $\dot{\boldsymbol{P}}_d$ corresponding to the force direction are equal to zero. Substituting $\boldsymbol{F} = \boldsymbol{F}_d$ in (3) and making use of constant $\boldsymbol{P}_e$ in the force direction, $\boldsymbol{P}_d$ is derived as

$$\boldsymbol{P}_d = \boldsymbol{P}_e - \boldsymbol{F}_d/\boldsymbol{K}_e. \tag{4}$$

Remark 1: In the grinding task, the robot end-effector exerts a force on the environment and moves perpendicular to the force direction. By making use of constant $\boldsymbol{P}_e$ in the force direction, the component in $\dot{\boldsymbol{P}}_e$ corresponding to the force direction is equal to zero. In other directions without force target, the components in $\boldsymbol{F}_d$ are set to zero, the components in $\boldsymbol{P}_e$ are determined to be equal to those of $\boldsymbol{P}_d$ according to (4). The motion objectives, such as position and velocity targets, can be set in those directions. It can be concluded that under grinding task, $\dot{\boldsymbol{P}}_e = \dot{\boldsymbol{P}}_d$ holds true in any direction.

Hence, the desired trajectory profile is decided by the environment surface location $\boldsymbol{P}_e$, environment stiffness $\boldsymbol{K}_e$, and the desired external force $\boldsymbol{F}_d$ to eliminate the state tracking errors. The workpiece measurement process is indispensable before machining. We assume that the environmental surface location is known beforehand. Then, we focus on the real-time estimation of the environment stiffness to calculate the precise new commanded desired trajectory.

Remark 2: The robot contact environment, clamped thin-walled workpiece, is characterized by low and variable stiffness, thus, vulnerable to deformation and vibration. Changes in environmental impedance can hinder the expected performance of the robot controller with fixed impedance. An appropriate adjustment of the impedance parameters of the robot can result

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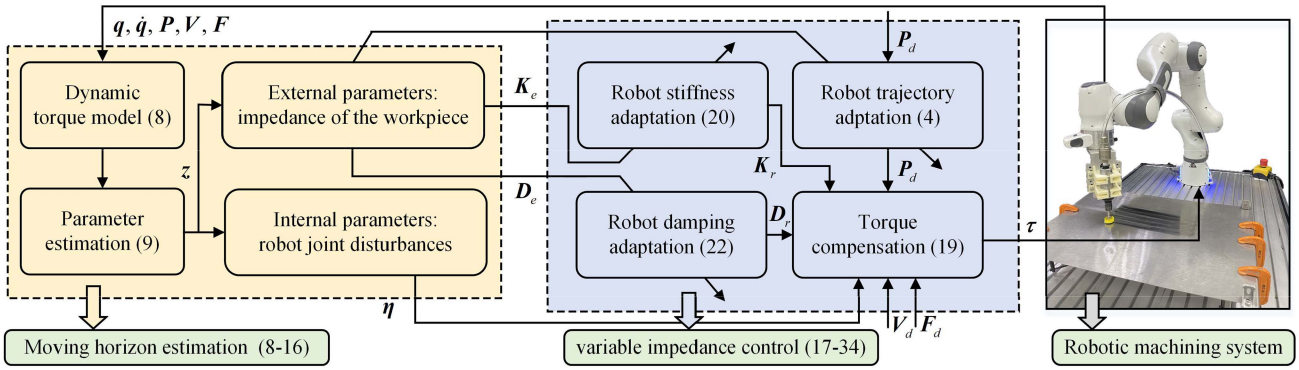


Fig. 3. Control block diagram of the proposed MHE-VIC scheme.

in an enhanced response to the impedance variations of the environment, leading to a more compliant contact behavior.

Thus, we proposed a moving horizon estimation-based variable impedance controller named MHE-VIC, the strategy of which is shown in Fig. 2(b). Define $\boldsymbol{y} = [\boldsymbol{P}, \boldsymbol{V}, \boldsymbol{F}, \boldsymbol{\eta}]^{\mathrm{T}}$ as the output variable, where $\boldsymbol{\eta} \in \mathbb{R}^m$ is the internal disturbance of the robot. At each control cycle t_k, the N historical measurements $\boldsymbol{y}(t_{k-N+1}), \ldots, \boldsymbol{y}(t_{k-1}), \boldsymbol{y}(t_k)$ are utilized to calculate the current unmeasurable useful parameters $\boldsymbol{z}(t_k) = [\boldsymbol{K}_e, \boldsymbol{D}_e, \boldsymbol{\eta}]^{\mathrm{T}}$ in the MHE scheme. $\boldsymbol{z}(t_k)$ contains the impedances of the external environment and the internal disturbance of the robot. Then, we designed a variable impedance controller using the online observations to improve robot's adaptability to the contact environment. The MHE-VIC method not only eliminates the state tracking errors in (1) but also perceives the environment impedance and regulates robot impedance to be more adaptable to the environment for accomplishing force-interacting tasks.

III. Observer-Based Variable Impedance Control

An observer-based variable impedance control approach, which unifies the MHE observer and the VIC controller, is developed in this section. The block diagram representing the transmission of internal data for the MHE-VIC scheme is depicted in Fig. 3. The MHE part estimates the environment impedance and robot disturbances, while the VIC part utilizes the observations to regulate robot impedance for improving the interaction performance. Specific discussions of MHE and VIC parts are, respectively, presented in the following sections.

A. Moving Horizon Based Robot-Environment Observer

The dynamics of the robot's physical interaction with the environment in Cartesian space are described as follows:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{P}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{P}} + \boldsymbol{G}(\boldsymbol{q}) + \boldsymbol{\eta} = \boldsymbol{u} + \boldsymbol{F} \tag{5}$$

where

$$\begin{cases} \boldsymbol{M}(\boldsymbol{q}) = \boldsymbol{J}^{-\mathrm{T}}(\boldsymbol{q})\boldsymbol{M}_I(\boldsymbol{q})\boldsymbol{J}^{-1}(\boldsymbol{q}) \\ \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{J}^{-\mathrm{T}}(\boldsymbol{q})(\boldsymbol{C}_I(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \boldsymbol{M}_I(\boldsymbol{q})\boldsymbol{J}^{-1}(\boldsymbol{q})\dot{\boldsymbol{J}}(\boldsymbol{q}))\boldsymbol{J}^{-1}(\boldsymbol{q}) \\ \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{J}^{-\mathrm{T}}(\boldsymbol{q})\boldsymbol{G}_I(\boldsymbol{q}). \end{cases} \tag{6}$$

$\boldsymbol{q} \in \mathbb{R}^n$ denotes the robot joint position. n is the number of the robot joints. $\boldsymbol{u} \in \mathbb{R}^m$ denotes the control input. $\boldsymbol{J} \in \mathbb{R}^{m \times n}$ is the Jacobian matrix of the robot. $\boldsymbol{M}_I(\boldsymbol{q}) \in \mathbb{R}^{n \times n}$ is the inertial matrix, $\boldsymbol{C}_I(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n \times n}$ is the matrix accounting for Coriolis, and centrifugal forces, $\boldsymbol{G}_I(\boldsymbol{q}) \in \mathbb{R}^n$ is the gravitational force.

Assuming the environment parameters $\boldsymbol{P}_e, \boldsymbol{K}_e$, and $\boldsymbol{D}_e$ vary slowly in a short neighboring control period (1 ms), we have $\boldsymbol{P}_e(t_k) = \boldsymbol{P}_e(t_{k-1})$, $\boldsymbol{K}_e(t_k) = \boldsymbol{K}_e(t_{k-1})$, and $\boldsymbol{D}_e(t_k) = \boldsymbol{D}_e(t_{k-1})$. Subtracting the force value at time t_{k-1} from the force value at time t_k according to (3) and simplifying $\boldsymbol{K}_e(t_k)$ to $\boldsymbol{K}_e$, it can be derived that

$$\begin{aligned} \boldsymbol{F}(t_k) - \boldsymbol{F}(t_{k-1}) = \boldsymbol{K}_e(\boldsymbol{P}(t_{k-1}) - \boldsymbol{P}(t_k)) \\ + \boldsymbol{D}_e(\dot{\boldsymbol{P}}(t_{k-1}) - \dot{\boldsymbol{P}}(t_k)). \end{aligned} \tag{7}$$

Let $\boldsymbol{x} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3]^{\mathrm{T}}$ be the system state variable, where $\boldsymbol{x}_1 = \boldsymbol{P}$, $\boldsymbol{x}_2 = \boldsymbol{V}$, and $\boldsymbol{x}_3 = \boldsymbol{F}$. Define $\boldsymbol{v} = \dot{\boldsymbol{x}}_2$ as an auxiliary variable. Based on (5)–(7), the state-space model of the robot-environment interaction system $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{z})$, can be established as

$$\begin{pmatrix} \dot{\boldsymbol{x}}_1 \\ \dot{\boldsymbol{x}}_2 \\ \dot{\boldsymbol{x}}_3 \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_2 \\ \boldsymbol{M}^{-1}(\boldsymbol{q})(\boldsymbol{u} - \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\boldsymbol{x}_2 - \boldsymbol{G}(\boldsymbol{q}) - \boldsymbol{x}_3 - \boldsymbol{\eta}) \\ \boldsymbol{K}_e\boldsymbol{x}_2 + \boldsymbol{D}_e\boldsymbol{v} \end{pmatrix}. \tag{8}$$

The impedance parameters of the thin-walled workpiece are hard to obtain. It is not accurate to calculate the impedance parameters of the workpiece by offline methods due to stochastic factors. It is preferable to estimate the impedance parameters in real time based on the observable force and position information. Based on (8), the moving horizon based robot-environment observer is designed as follows:

$$\begin{aligned} \min_{\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{z}} \Bigg\{ & \left\| \begin{matrix} \boldsymbol{x}(t_{k-N+1}) - \bar{\boldsymbol{x}} \\ \boldsymbol{z} - \bar{\boldsymbol{z}} \end{matrix} \right\|_{\bar{\boldsymbol{Q}}}^2 \\ & + \sum_{i=k-N+1}^{k} \|(\boldsymbol{y}_r(t_i) - \boldsymbol{h}(\hat{\boldsymbol{x}}(t_i), \hat{\boldsymbol{u}}(t_i), \hat{\boldsymbol{v}}(t_i), \hat{\boldsymbol{z}})\|_{\boldsymbol{R}_i}^2 \Bigg\} \end{aligned} \tag{9}$$

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subjected to

$$\left.\begin{aligned}&\dot{\hat{\boldsymbol{x}}}(t_s)=f(\hat{\boldsymbol{x}}(t_s),\hat{\boldsymbol{u}}(t_s),\hat{\boldsymbol{v}}(t_s),\hat{\boldsymbol{z}})\\&\hat{\boldsymbol{y}}(t_s)=h(\hat{\boldsymbol{x}}(t_s),\hat{\boldsymbol{u}}(t_s),\hat{\boldsymbol{v}}(t_s),\hat{\boldsymbol{z}})\\&\hat{\boldsymbol{x}}(t_s)\in\chi,\hat{\boldsymbol{u}}(t_s)\in\mu,\hat{\boldsymbol{v}}(t_s)\in\upsilon,\hat{\boldsymbol{z}}\in\rho\end{aligned}\right\}t_s\in[t_i,t_{i+1}]\quad(10)$$

where $(\bar{\boldsymbol{x}},\bar{\boldsymbol{z}},\bar{\boldsymbol{Q}})$ is the initial regulation data of state, estimation vector, and coefficients at t_{k-N+1}, the values are obtained by the estimated results using the extended Kalman filter (EKF) update [43] in the last optimization procedure. For a vector $\boldsymbol{a}$ and matrix $\boldsymbol{A}$, $\|\boldsymbol{a}\|_{\boldsymbol{A}}^2=\boldsymbol{a}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{a}$, and $\hat{\boldsymbol{a}}$ represents the estimated value of $\boldsymbol{a}$. The total time of N control periods is defined as the time horizon $T_E=t_k-t_{k-N+1}$. $\boldsymbol{R}_i\in\mathbb{R}^{4m\times4m}$ is a diagonal matrix with positive entries, which is set to the inverse of the covariance matrix for measurement noise. $\boldsymbol{y}_r$ is the measured historical reference of the estimated output variable $\hat{\boldsymbol{y}}=[\hat{\boldsymbol{P}},\hat{\boldsymbol{V}},\hat{\boldsymbol{F}},\hat{\boldsymbol{\eta}}]^{\mathrm{T}}$. $\hat{\boldsymbol{\eta}}$ is the estimated disturbance term of the robot in the Cartesian space, the measurement value of which can be constructed from robot joint torque

$$\boldsymbol{\eta}_r=\boldsymbol{J}^{-\mathrm{T}}\boldsymbol{\tau}_\eta=\boldsymbol{J}^{-\mathrm{T}}\boldsymbol{\tau}_d-\boldsymbol{u}\quad(11)$$

$$\boldsymbol{\tau}_d=\boldsymbol{J}^{\mathrm{T}}(\boldsymbol{M}(\boldsymbol{P}_d)\ddot{\boldsymbol{P}}_d+\boldsymbol{C}(\boldsymbol{P}_d,\dot{\boldsymbol{P}}_d)\dot{\boldsymbol{P}}_d+\boldsymbol{G}(\boldsymbol{P}_d)-\boldsymbol{F})\quad(12)$$

where $\boldsymbol{\tau}_\eta\in\mathbb{R}^n$ denotes the disturbance term in the joint space. $\boldsymbol{\tau}_d\in\mathbb{R}^n$ is the desired torque when the robot executes the desired trajectory suffering the current interaction force. The precise form of $\boldsymbol{\tau}_d$ is given in (12). It is worth noting that the robot torque controller needs precise torque values for accurate movement. A little torque deviation will lead to completely different action responses at the robot end-effector. The torque input $\boldsymbol{u}$ without compensating disturbance $\boldsymbol{\eta}$ will decrease the controller performance. The constraints of the optimization problem (9) in (10) are set as follows:

$$\begin{cases}\chi=\{\hat{\boldsymbol{x}}\in\mathbb{R}^{3m}|\,\|\hat{\boldsymbol{x}}\|\le\boldsymbol{x}_r+\Delta x_{\max},\left\|\dot{\hat{\boldsymbol{x}}}\right\|\le\Delta\dot{x}_{\max}\}&(13)\\\mu=\{\hat{\boldsymbol{u}}\in\mathbb{R}^m|\,\|\hat{\boldsymbol{u}}\|\le\boldsymbol{u}_r+\boldsymbol{J}^{-\mathrm{T}}\Delta\boldsymbol{\tau}_{\max}\}&(14)\\\upsilon=\{\hat{\boldsymbol{v}}\in\mathbb{R}^m|\,\|\hat{\boldsymbol{v}}\|\le\boldsymbol{v}_{\max}\}&(15)\\\rho=\{\hat{\boldsymbol{z}}\in\mathbb{R}^{(2m+1)\times m}|\,\|\hat{\boldsymbol{z}}\|\le\boldsymbol{z}_{\max}\}&(16)\end{cases}$$

where $\boldsymbol{x}_r$ and $\boldsymbol{u}_r$ denote the measured values of $\boldsymbol{x}$ and $\boldsymbol{u}$, symbol $\Delta(\cdot)$ denotes the increment of the variable in brackets, and the subscript $(\cdot)_{\max}$ represents the maximum value of the variable in brackets. Compared with other estimation strategies, the physical constraints of MHE can improve the accuracy of estimation results. It should be mentioned that the proposed robot-environment observer was designed using the MHE scheme outlined in [36], and a detailed stability analysis of the observer within this scheme was presented in [37]. In our case, we are able to simplify the system model to be linear using feedback linearization technology, resulting in a stable estimator of (9) through the EKF update. Based on (9)–(16), the customized MHE of the robot-environment observer is formulated. The observations will, then, be used to regulate the robot's control inputs for better force interaction performance.

B. Variable Impedance Controller Design

In this section, we aim to develop a variable impedance strategy to enhance the robot's adaptability to its contact environment by regulating its impedance in accordance with the estimated environment impedance, and to eliminate state tracking errors. Combining (2) and (5), the robot control input is designed as

$$\boldsymbol{u}=\boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}+\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\boldsymbol{x}_2+\boldsymbol{G}(\boldsymbol{q})+\boldsymbol{\eta}-\boldsymbol{x}_3\quad(17)$$

where

$$\begin{aligned}\boldsymbol{v}=\ddot{\boldsymbol{P}}_d+\boldsymbol{M}_r^{-1}(-\boldsymbol{D}_r(t_k)(\dot{\boldsymbol{P}}-\dot{\boldsymbol{P}}_d)\\-\boldsymbol{K}_r(t_k)(\boldsymbol{P}-\boldsymbol{P}_d)+\boldsymbol{F}-\boldsymbol{F}_d).\end{aligned}\quad(18)$$

$\boldsymbol{M}_r$ is set to a constant diagonal matrix. $\boldsymbol{K}_r$ and $\boldsymbol{D}_r$ are designed as variable diagonal matrices, which are represented as $\boldsymbol{K}_r(t_k)$ and $\boldsymbol{D}_r(t_k)$ in the remaining parts. For simplicity, the time variable is omitted when describing other variables. $\boldsymbol{u}$ is defined in the Cartesian space and should be converted to joint torque $\boldsymbol{\tau}\in\mathbb{R}^n$ to control the real robot

$$\begin{aligned}\boldsymbol{\tau}&=\boldsymbol{J}^{\mathrm{T}}\boldsymbol{u}\\&=\boldsymbol{J}^{\mathrm{T}}(\boldsymbol{M}(\boldsymbol{q})(\ddot{\boldsymbol{P}}_d+\boldsymbol{M}_r^{-1}(-\boldsymbol{D}_r(t_k)(\boldsymbol{x}_2-\dot{\boldsymbol{P}}_d)\\&\quad-\boldsymbol{K}_r(t_k)(\boldsymbol{x}_1-\boldsymbol{P}_d)\\&\quad+\boldsymbol{F}-\boldsymbol{F}_d))+\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\boldsymbol{x}_2+\boldsymbol{G}(\boldsymbol{q})+\boldsymbol{\eta}-\boldsymbol{x}_3).\end{aligned}\quad(19)$$

Theorem 1: Considering robot dynamics (5), the robot force tracking impedance model (2), and the impedance model of the contact environment (3), the variable impedance control law (19) will eliminate the state tracking errors in (1) and ensure the asymptotic stability of the robot-environment interaction system if the robot stiffness $\boldsymbol{K}_r(t_k)$ and damping $\boldsymbol{D}_r(t_k)$ are treated as continuously varying matrices and selected as

$$\begin{cases}\boldsymbol{K}_r(t_k)=\hat{\boldsymbol{K}}_e(t_k)&(20)\\\dot{\hat{\boldsymbol{K}}}_e(t_k)<\hat{\boldsymbol{K}}_e(t_k)&(21)\\\boldsymbol{D}_r(t_k)=\boldsymbol{D}^*-\hat{\boldsymbol{D}}_e(t_k)&(22)\\\boldsymbol{D}^*>\boldsymbol{M}_r/2&(23)\end{cases}$$

where $\boldsymbol{D}^*\in\mathbb{R}^{m\times m}$ is a customized damping matrix. Robot impedance varies with the estimated environmental impedance to improve the interaction performance resulting in reduced vibration in the machining task. Assuming that the estimated environmental stiffness is continuous, the rate of change of estimated environmental stiffness can be controlled by the robot's velocity.

Proof: According to (4), we have $\boldsymbol{F}_d=\boldsymbol{K}_e(\boldsymbol{P}_e-\boldsymbol{P}_d)$. Using $\boldsymbol{e}_F=\boldsymbol{F}-\boldsymbol{F}_d$ to denote the force tracking error and removing the term $\boldsymbol{F}_d$ in (4) using $\boldsymbol{F}_d=\boldsymbol{F}-\boldsymbol{e}_F$, we have

$$\boldsymbol{e}_F=\boldsymbol{F}-\boldsymbol{K}_e(\boldsymbol{P}_e-\boldsymbol{P}_d).\quad(24)$$

Substituting $\boldsymbol{F}=\boldsymbol{K}_e(\boldsymbol{P}_e-\boldsymbol{P})+\boldsymbol{D}_e(\dot{\boldsymbol{P}}_e-\dot{\boldsymbol{P}})$ in (24), we have

$$\boldsymbol{e}_F=\boldsymbol{K}_e(\boldsymbol{P}_d-\boldsymbol{P})+\boldsymbol{D}_e(\dot{\boldsymbol{P}}_e-\dot{\boldsymbol{P}}).\quad(25)$$

Using $\boldsymbol{e}_P=\boldsymbol{P}-\boldsymbol{P}_d$ to denote the position tracking error. If $\boldsymbol{e}_p\to0$, we will have $\boldsymbol{P}\to\boldsymbol{P}_d$. According to remark 1, we have

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$\dot{\boldsymbol{P}}_e = \dot{\boldsymbol{P}}_d$. If $\dot{\boldsymbol{e}}_p \to 0$, we will have $\dot{\boldsymbol{P}} \to \dot{\boldsymbol{P}}_d$, thus, $\dot{\boldsymbol{P}} \to \dot{\boldsymbol{P}}_e$. Based on the abovementioned analysis, the force tacking error $\boldsymbol{e}_F$ in (25) will converge to zero if $\boldsymbol{e}_p$ and $\dot{\boldsymbol{e}}_p$ converge to zero.

Substituting (3) into (2), removing $\boldsymbol{F}_d$ in (2) based on (4), and replacing the real impedance variables with the estimated impedance variables, the combined impedance model of the robot-environment interaction system can be derived as

$$\begin{aligned}&\boldsymbol{M}_r(\ddot{\boldsymbol{P}} - \ddot{\boldsymbol{P}}_d) + \boldsymbol{D}_r(t_k)(\dot{\boldsymbol{P}} - \dot{\boldsymbol{P}}_d) + \boldsymbol{K}_r(t_k)(\boldsymbol{P} - \boldsymbol{P}_d)\\ &\quad = \hat{\boldsymbol{K}}_e(\boldsymbol{P}_e - \boldsymbol{P}) + \hat{\boldsymbol{D}}_e(\dot{\boldsymbol{P}}_e - \dot{\boldsymbol{P}}) - \hat{\boldsymbol{K}}_e(\boldsymbol{P}_e - \boldsymbol{P}_d). \quad (26)\end{aligned}$$

Removing $\dot{\boldsymbol{P}}_d$ in (26) as $\dot{\boldsymbol{P}}_e$, we have

$$\boldsymbol{M}_r\ddot{\boldsymbol{e}}_p + (\boldsymbol{D}_r(t_k) + \hat{\boldsymbol{D}}_e(t_k))\dot{\boldsymbol{e}}_p + (\boldsymbol{K}_r(t_k) + \hat{\boldsymbol{K}}_e(t_k))\boldsymbol{e}_p = 0. \quad (27)$$

If $\boldsymbol{e}_p \to 0$ and $\dot{\boldsymbol{e}}_p \to 0$, then $\ddot{\boldsymbol{e}}_p$ will also tend to zero. Let $\boldsymbol{K}^*$ and $\boldsymbol{D}^* \in \mathbb{R}^{m\times m}$ be defined as the aggregate stiffness matrix and aggregate damping matrix, respectively, as specified in

$$\begin{cases}\boldsymbol{K}^* = \boldsymbol{K}_r(t_k) + \hat{\boldsymbol{K}}_e(t_k) & (28)\\ \boldsymbol{D}^* = \boldsymbol{D}_r(t_k) + \hat{\boldsymbol{D}}_e(t_k). & (29)\end{cases}$$

The subsequent stability analysis is predominantly based on [26]. The Lyapunov function of (27) is constructed as follows:

$$V_L = \frac{(2\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p)^{\mathrm{T}}\boldsymbol{M}_r(2\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p)}{2} + \frac{\boldsymbol{e}_p^{\mathrm{T}}\boldsymbol{\beta}\boldsymbol{e}_p}{2} \quad (30)$$

where $\boldsymbol{\beta} = \boldsymbol{K}^* + \boldsymbol{D}^*/2 - \boldsymbol{M}_r/4$. Function $V_L \geq 0$, and $V_L = 0$ only when $\boldsymbol{e}_p$ and $\dot{\boldsymbol{e}}_p$ are equal to zero. The derivative of (30) is continuous and can be derived as

$$\begin{aligned}\dot{V}_L = {}& \dot{\boldsymbol{e}}_p^{\mathrm{T}}(\boldsymbol{M}_r/2 - \boldsymbol{D}^*)\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}(\boldsymbol{M}_r/4\\ &+ \boldsymbol{\beta} - \boldsymbol{K}^* - \boldsymbol{D}^*/2)\dot{\boldsymbol{e}}_p\\ &+ \boldsymbol{e}_p^{\mathrm{T}}(2\dot{\boldsymbol{\beta}} - \boldsymbol{K}^*)\boldsymbol{e}_p\Big/4. \quad (31)\end{aligned}$$

Since $\boldsymbol{M}_r$ is set to a constant diagonal matrix, we have $\dot{\boldsymbol{\beta}} = \dot{\boldsymbol{K}}^* + \dot{\boldsymbol{D}}^*/2$. Removing $\boldsymbol{\beta}$ and $\dot{\boldsymbol{\beta}}$ in (31), we have

$$\dot{V}_L = \dot{\boldsymbol{e}}_p^{\mathrm{T}}(\boldsymbol{M}_r/2 - \boldsymbol{D}^*)\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}((2\dot{\boldsymbol{K}}^* + \dot{\boldsymbol{D}}^*) - 2\boldsymbol{K}^*)\boldsymbol{e}_p\Big/4. \quad (32)$$

In order to make $\dot{V}_L \leq 0$, the following equations should be satisfied:

$$\begin{cases}\boldsymbol{M}_r < 2\boldsymbol{D}^* & (33)\\ (2\dot{\boldsymbol{K}}^* + \dot{\boldsymbol{D}}^*) < 2\boldsymbol{K}^*. & (34)\end{cases}$$

The impedance selection conditions of (20)–(23) satisfy (33) and (34). Thus, the derivative of the Lyapunov function will be negative when $\boldsymbol{e}_p \neq 0$ and $\dot{\boldsymbol{e}}_p \neq 0$. According to Lyapunov's second theorem, the robot-environment interaction system is stable, and the state tracking errors in (1) converge to zero. The analysis of the Lyapunov function and the derivation details of (31) are given in Appendix A.

IV. Experimental Results

In order to validate the efficacy of the MHE-VIC method, thin-walled workpiece grinding experiments are carried out on a panda robot. Fig. 4 displays the hardware of the robot, machining tools, and sensors for machining experiments. The robot and workpiece data, including position, velocity, force, and vibration, are collected. The thin-walled workpiece is fixed by four steel pillars at the four corners. We use contact environment to describe the thin-walled workpiece clamped in the above way. The contact stiffness value is different in the different positions of the workpiece. In the experiment, the Panda robot is commanded in a pure joint torque mode within a 1 ms control cycle. The software framework is built in the Ubuntu system under the robot operating system (ROS). The ACADO toolkit [44] is used to generate a real-time MHE profile, which is integrated into the ROS-control interface for coding the rest VIC part.

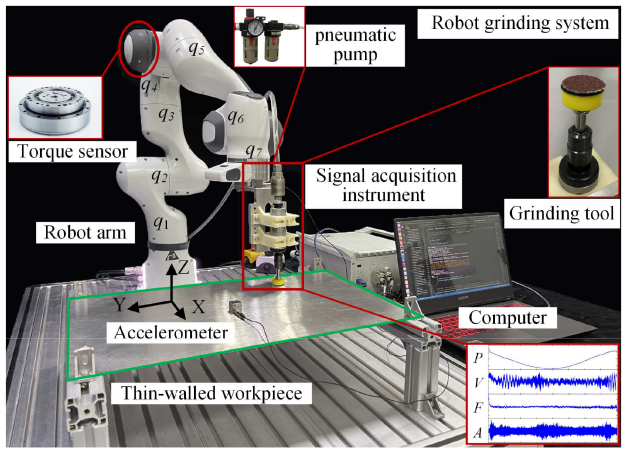


Fig. 4. Experimental scene of robot grinding thin-walled workpiece.

A. Performance Verification of MHE-VIC Method

The thickness of the thin-walled workpiece in the experiment is merely 0.68 mm. The machining robot maintained a uniform linear motion of 20 mm/s in the *Y*-direction for 16 s and exerted a constant external force of 8 N in the *Z*-direction to grind the workpiece. Impedance parameters in the *Z*-direction of the workpiece were observed in real-time during the experiment. Unbolded symbols represent data in one dimension. The robot's posture, velocity, and force states can be fetched in the feedback procedure for each 1 ms. The control instant is set to 1 ms, and the moving horizon is set to 20 ms. The Gauss–Newton Hessian matrix approximation method is used to solve the second-order partial derivatives of the Lagrange function of the nonlinear optimization problem. The multiple-shooting method is adopted for nonlinear equation discretization. The Levenberg–Marquardt algorithm is adopted for searching solutions. The absolute tolerance of the integrator used to solve integral terms in (9) is set to 10^{-10}. The specific parameter settings of the robotic machine system are given in Table I.

The robot initial state variable $\boldsymbol{x}(t_0) = [\boldsymbol{P}(t_0), \boldsymbol{V}(t_0), \boldsymbol{F}(t_0)]^{\mathrm{T}}$, where position $\boldsymbol{P}(t_0) = [0.611\mathrm{m}, 0.184\mathrm{m}, 0.135\mathrm{m}]^{\mathrm{T}}$, velocity $\boldsymbol{V}(t_0) = \boldsymbol{0}$, and force $\boldsymbol{F}(t_0) = \boldsymbol{0}$. The corresponding joint position of the robot is $\boldsymbol{q}(t_0) = [0.178, -0.056, 0, -2.262,$

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TABLE I
PARAMETER SETTING OF THE MACHINE SYSTEM AND ROBOT CONTROLLER

Thin-walled workpiece (304 steel plate)	1) Length: 400 mm 2) Width: 350 mm 3) Height: 0.68 mm
Pneumatic grinding tool (Ingersoll Rand 307)	1) Air pressure: 0.6 MPa 2) Grinding speed: 28000 r/min 3) Diameter: 25mm
Moving horizon estimation coefficients	1) Historical horizon $T_E = 20$ ms 2) Initial value of $\boldsymbol{R}_i$ and $\bar{\boldsymbol{Q}}$: $\boldsymbol{R}_i = \text{diag}(10^5, 10^3, 10^2, 10^2)$ $\bar{\boldsymbol{Q}} = \text{diag}(10^5, 10^3, 10^2, 10^2, 10, 10^2)$ 3) $\triangle \boldsymbol{x}_{max} = [\triangle P, \triangle V, \triangle F]$, $\triangle P = 1$ mm $\triangle V = 5$ mm/s, $\triangle F = 1$ N, $\triangle \dot{P} = 5$ mm/s $\triangle \dot{V} = 20$ mm/s^2, $\triangle \dot{F} = 5$ N/s 4) $\triangle \tau_{\max} = 1$ N·m, $\boldsymbol{v}_{\max} = 1$·m/s^2 5) $\boldsymbol{z}_{max} = [K_{max}, D_{max}, \eta_{max}]^{\mathrm{T}}$, $K_{max} = 10^4$ N/m $D_{max} = 200$ N/(m/s), $\eta_{max} = 5$ N
Variable impedance control coefficients	1) Sampling time T: 1 ms 2) Normal grinding force F^d : 8 N 3) Tangential feed speed V^d : 20 mm/s

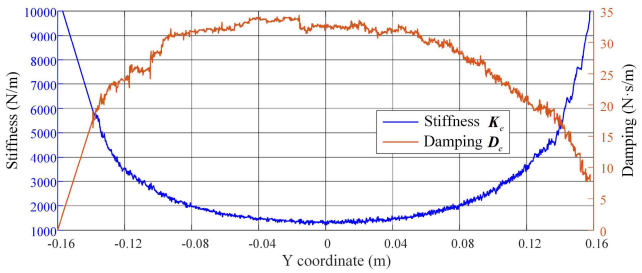


Fig. 5. Estimated stiffness (blue) and damping (orange) parameters of the external contact environment in the Z direction.

0.001, 2.322, 0.191]^{T}. The unit of joint angles is radians. The real-time dynamic model parameters of the robot are taken from the internal model library of the Franka Panda robot. When the robot executes the grinding trajectory, the MHE part observes the environment impedance and the robot joint disturbances. The main force interaction occurred in the Z-direction, and we processed the data in this direction for simplicity. The estimated stiffness and damping parameters of the contact environment are depicted by blue and orange lines in Fig. 5, respectively. It can be seen that the estimated stiffness varies from 1200 N/m to 10 000 N/m. The lowest stiffness is in the middle of the workpiece. The change in the damping parameter is opposite to the stiffness parameter. The maximum damping parameter appears in the middle of the workpiece. The robot impedance varies with the environmental impedance according to (20) and (22) for interacting with the environment adaptively. The observation of the robot's internal disturbance in the Z-direction, η_z, is shown in Fig. 6. According to (11), disturbances in Cartesian space can be transformed into joint space, which is shown in Fig. 7. The robot end-effector is perpendicular to the horizontal plane. Its trajectory determines that only joint 2, joint 4, and joint 6 affect the Z-direction motion accuracy. Fig. 7(a) shows joint disturbance torques in joint 2, joint 4, and joint 6. Fig. 7(b) depicts the joint torques with disturbance and with disturbance

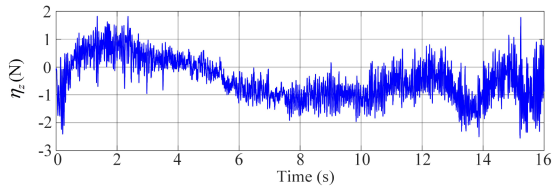


Fig. 6. Estimated robot internal disturbance in the Z-direction.

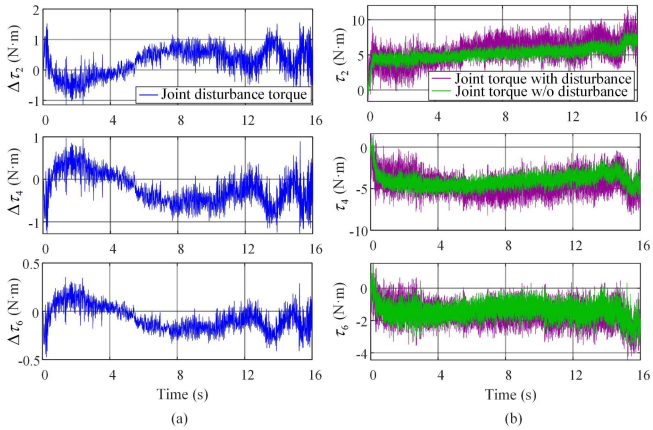


Fig. 7. (a) Joint disturbance torques in joint 2, joint 4, and joint 6, respectively. (b) Joint torque input values in joint 2, joint 4, and joint 6 with disturbances (purple) and with disturbances compensation (green).

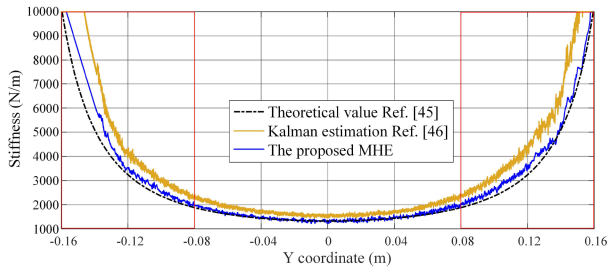


Fig. 8. Theoretical workpiece stiffness, Kalman estimation stiffness, and the proposed MHE stiffness.

compensation in purple and green lines, respectively. It is obvious that the disturbances affect the value of joint torque inputs. It should be noted that a little difference in the joint torque of the robot will lead to a completely different response. Thus, the compensation for the disturbances is indispensable, and the specific impact can be reflected in the control results. As nominal values for damping and robot disturbance are unavailable, we only compare the estimated stiffness value with its theoretical value [45] and Kalman estimation value [46]. As shown in Fig. 8, the blue line is closer to the black dotted line than the orange line, which means the accuracy of the proposed MHE is better than that of the Kalman method. The average stiffness error of the Kalman method is 570 N/m, while the average stiffness error of the proposed MHE is only 190 N/m. The red boxes in Fig. 8 correspond to areas of high workpiece stiffness. The machining vibrations caused by force interaction are stronger in

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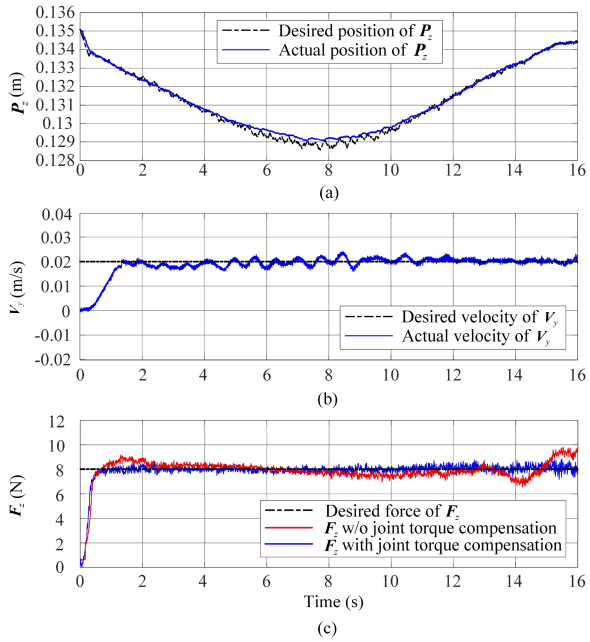


Fig. 9. (a) Desired and actual position of the robot end-effector in the *Z*-direction. (b) Desired and actual velocity of the robot end- effector in the *Y*-direction. (c) Black dotted line depicts the desired external force of the robot in the *Z*-direction. The actual external force of the robot without disturbances compensation and with disturbances compensation are represented in the red and blue lines, respectively.

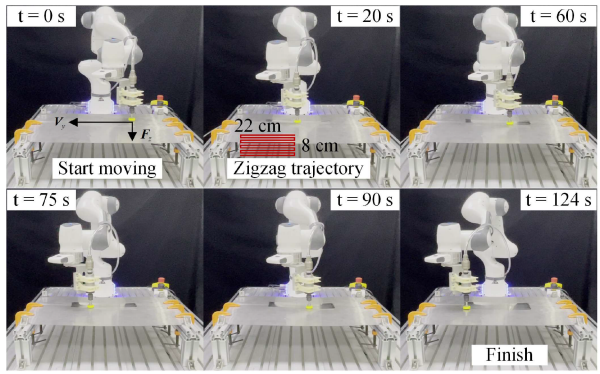


Fig. 10. Moving states for robot grinding the central region of the thin-walled workpiece through the zigzag trajectory.

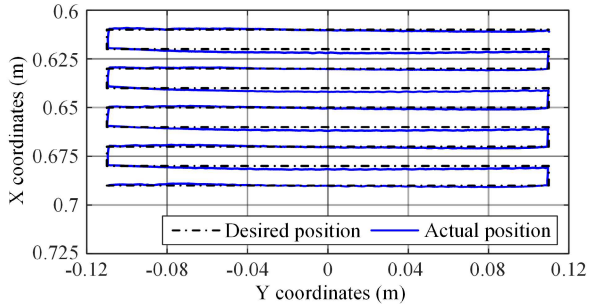


Fig. 11. Zigzag trajectory tracking performance in the X-Y plane.

these regions, while the constraint properties of MHE make it still maintain accurate estimation performance.

According to the proposed VIC (19), the state tracking performance of (1) is shown in Fig. 9. The blue lines depict the actual values of the state variables, while the dotted black lines are the desired values of the corresponding variable. The results in Fig. 9(a) show that the desired position of the robot end-effector is changing according to (4), and the position tracking error is less than 0.5 mm in the *Z*-direction. Fig. 9(b) demonstrates that the velocity tracking error is less than 0.005 m/s in the *Y*-direction. Fig. 9(c) describes the force tracking results of the VIC method with disturbance compensation and without disturbance compensation. The comparison between the blue and red lines shows that there will be an obvious force error without disturbance compensation in the *Z*-direction. The proposed method with disturbance compensation can ensure the convergence of the force tracking error, and the force tracking accuracy is less than 1 N.

B. Machining Quality Analysis for Different Methods

In order to verify the machining quality of the proposed method, the robot is coded to grind the central region on the workpiece by using a zigzag trajectory. The zigzag trajectory is widely used in machining tasks to cover the entire surface of the workpiece effectively. As shown in Fig. 10, the grinding region is 22 cm in length with 8 cm in width. The robot started from $\boldsymbol{P}(t_0) = [0.611\text{m}, 0.110\text{m}, 0.135\text{m}]^{\text{T}}$, and moved in the *Y*-direction by accelerating at -0.02 m/s^2 for 1 s, then maintained a speed of -0.02 m/s for 10 s, and finally decelerated at 0.02 m/s^2 for 1 s. In the *X*-direction, the robot accelerated at 0.01 m/s^2 for 1 s, then decelerated at -0.01 m/s^2 for 1 s. This was repeated continuously to cover the entire grinding area. In the *Z*-direction, the desired position was set to $0.135 - 8/K_e(t_k)$. The whole process consumed 124 s. The desired grinding trajectory and the tracking performance in the X-Y plane are enlarged and displayed in Fig. 11. The position accuracy of the whole process is better than 0.5 mm. The machining quality is shown in Fig. 12. It can be found that the grinding area with the proposed controller is almost mirror-like. The reflection of the robot and end-effector can be easily seen in the control-ON region but can hardly be seen in the control-OFF region.

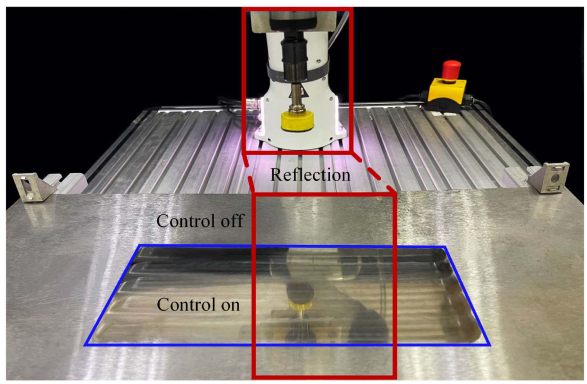


Fig. 12. Image of the thin-walled workpiece before and after grinding.

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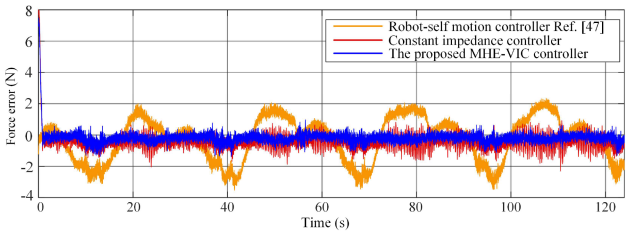


Fig. 13. Force tracking results of the robot motion controller, constant impedance controller, and the proposed MHE-VIC controller.

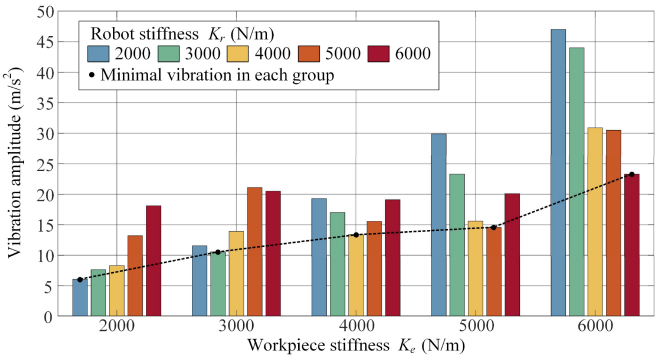


Fig. 14. Workpiece vibration results with different robot stiffnesses during machining thin-walled workpieces.

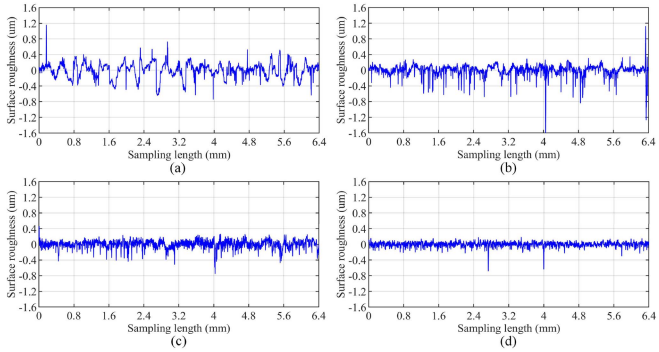


Fig. 15. Surface roughness of the workpiece. (a) Region before grinding. (b) Region after grinding by the robot motion controller. (c) Region after grinding by constant impedance controller. (d) Region after grinding by the proposed MHE-VIC controller.

TABLE II
SURFACE ROUGHNESS RESULTS OF DIFFERENT GRINDING METHODS

Index	Unmachined workpiece	Motion control	Constant impedance	Variable impedance
R_a (μm)	0.150	0.090	0.076	0.042
R_z (μm)	1.103	1.211	0.714	0.479

To further demonstrate the superiority of the proposed method, the Panda robot-self motion controller [47] and impedance controller (2) with constant impedance parameters are adopted for conducting the same zigzag grinding task. The force tracking comparison results of three different methods are shown in Fig. 13. The orange line represents the robot motion controller result. It can maintain a high-level position accuracy by tracking the prescribed trajectory but performs worse in force tracking performance due to the low stiffness of the workpiece. The red line depicts the results of the constant impedance controller, which has a kind of ability to deal with the force error but still has some jitters because of the variable stiffness attributes of the contact environment. The force accuracy of the constant impedance controller is about 2 N. The proposed MHE-VIC method corresponds to the blue line. The force accuracy is less than 1 N, which is better than the previous two methods. The results of workpiece vibration are presented in Fig. 14. Vibration data was collected at different positions of the workpiece, where the workpiece impedance varied from 2000 N/m to 6000 N/m. The results demonstrate that reduced vibration is achieved when the robot impedance is set in proximity to the workpiece impedance. Conversely, when the robot impedance deviates from the workpiece impedance, the vibration increases. The minimal vibration amplitude occurs when the robot and workpiece impedance are equal, which demonstrates the effectiveness of the proposed variable impedance law (20). The abovementioned results show that the proposed variable impedance strategy effectively responds to impedance changes in the environment, leading to improved state tracking performance and reduced vibration behavior.

The surface roughness of the machined regions is measured by Mitutoyo Sj-210. The sampling length is 6.4 mm. Fig. 15(a)–(d) represents the surface roughness of the unmachined part, the machined part by the motion controller, the machined part by the constant impedance controller, and the machined part by the proposed method, respectively. It is obvious that the consistency of the surface roughness is getting better from Fig. 15(a)–(d). The results of average height deviations from the mean line and the maximum peak-to-valley height of the profile, *Ra* and *Rz*, are given in Table II. The *Ra* of the unmachined workpiece is 0.15 μm. The *Ra* of the machined workpiece of the constant impedance controller, 0.076 μm, is better than that of the motion controller, 0.09 μm. The *Ra* of the workpiece machined by the proposed method is 0.042 μm. The value of the *Ra* of the proposed variable impedance controller is increased by 72% more than the unmachined workpiece and 45% more than the constant impedance controller. The *Rz* results also show the superiority of the proposed controller. It can be seen that the motion controller even makes the *Rz* value worse than the unmachined workpiece. That demonstrates how important to apply compliant machining ways. Moreover, while the constant impedance method has compliant force behavior, the impedance changing of the environment will weaken this ability. The proposed MHE-VIC method can perceive the current environment impedance and regulate robot impedance to make the robot flexible enough to fit the changing environment for better machining quality.

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V. Conclusion

In this article, an observer-based variable impedance control approach was proposed to improve the robot adaptability of machining thin-walled workpieces. To obtain the unmeasurable parameters in the robot-environment system, a moving horizon based observer was developed to simultaneously perceive the impedance of the external environment and the disturbances of the robot's internal torque. To enhance the adaptability of the robot to the environment, a variable impedance controller was designed using observation results. The robot impedance varies with the environment impedance to eliminate the state tracking errors and reduce the machining vibration. The disturbance compensation procedure improves the controller's accuracy. To verify the efficacy of the proposed method, thin-walled workpiece grinding experiments were conducted. The results manifest that the accuracy of position, velocity, and force is higher than 0.5 mm, 0.005 m/s, and 1 N, respectively. The workpiece surface roughness further indicates the superiority of the proposed method to the robot motion controller and constant impedance controller.

Appendix A
Analysis of the Lyapunov Function and Its Derivative

The Lyapunov function is constructed as (30). The term $\boldsymbol{e}_p$ and $\dot{\boldsymbol{e}}_p$ are column vectors. $\boldsymbol{M}_r$ is the mass matrix of the impedance model (2), which is defined as a diagonal matrix with positive entries. $\boldsymbol{\beta} = \boldsymbol{K}^* + \boldsymbol{D}^*/2 - \boldsymbol{M}_r/4$, where $\boldsymbol{K}^*$ and $\boldsymbol{D}^*$ are also diagonal matrix with positive entries. The condition (33) is $\boldsymbol{M}_r < 2\boldsymbol{D}^*$, which indicates $\boldsymbol{D}^*/2 > \boldsymbol{M}_r/4$. Thus, $\boldsymbol{\beta}$ is a diagonal matrix with positive entries. In summary, for any $\boldsymbol{e}_p$ and $\dot{\boldsymbol{e}}_p$, Lyapunov function $V_L \geq 0$, and $V_L = 0$ only when $\boldsymbol{e}_p$ and $\dot{\boldsymbol{e}}_p$ are equal to zero.

According to (27), it can be derived that

$$\ddot{\boldsymbol{e}}_p = \frac{1}{-\boldsymbol{M}_r}(\boldsymbol{D}_r(t_k) + \hat{\boldsymbol{D}}_e(t_k))\dot{\boldsymbol{e}}_p + (\boldsymbol{K}_r(t_k) + \hat{\boldsymbol{K}}_e(t_k))\boldsymbol{e}_p. \tag{35}$$

The derivative of the Lyapunov function can be obtained in the following derivation by removing $\ddot{\boldsymbol{e}}_p$ using (35):

$$\begin{aligned}
\dot{V}_L &= (\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p/2)^{\mathrm{T}}\boldsymbol{M}_r(\ddot{\boldsymbol{e}}_p + \dot{\boldsymbol{e}}_p/2) + \boldsymbol{e}_p^{\mathrm{T}}\boldsymbol{\beta}\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}\dot{\boldsymbol{\beta}}\boldsymbol{e}_p\Big/2 \\
&= \dot{\boldsymbol{e}}_p^T\boldsymbol{M}_r\ddot{\boldsymbol{e}}_p + \dot{\boldsymbol{e}}_p^T\boldsymbol{M}_r\dot{\boldsymbol{e}}_p/2 + \boldsymbol{e}_p^{\mathrm{T}}\boldsymbol{M}_r\ddot{\boldsymbol{e}}_p/2 \\
&\quad + \boldsymbol{e}_p^{\mathrm{T}}(\boldsymbol{M}_r/4 + \boldsymbol{\beta})\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}\dot{\boldsymbol{\beta}}\boldsymbol{e}_p\Big/2 \\
&= [\dot{\boldsymbol{e}}_p^T + \boldsymbol{e}_p^{\mathrm{T}}/2][-(\boldsymbol{D}_r(t_k) + \hat{\boldsymbol{D}}_e(t_k))\dot{\boldsymbol{e}}_p \\
&\quad - (\boldsymbol{K}_r(t_k) + \hat{\boldsymbol{K}}_e(t_k))\boldsymbol{e}_p] \\
&\quad + \dot{\boldsymbol{e}}_p^T\boldsymbol{M}_r\dot{\boldsymbol{e}}_p/2 + \boldsymbol{e}_p^{\mathrm{T}}\boldsymbol{M}_r\dot{\boldsymbol{e}}_p/4 + \boldsymbol{e}_p^{\mathrm{T}}\boldsymbol{\beta}\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}\dot{\boldsymbol{\beta}}\boldsymbol{e}_p\Big/2 \\
&= -[\dot{\boldsymbol{e}}_p^T + \boldsymbol{e}_p^{\mathrm{T}}/2][\boldsymbol{D}^*\dot{\boldsymbol{e}}_p + \boldsymbol{K}^*\boldsymbol{e}_p] + \dot{\boldsymbol{e}}_p^T\boldsymbol{M}_r\dot{\boldsymbol{e}}_p/2 \\
&\quad + \boldsymbol{e}_p^{\mathrm{T}}(\boldsymbol{M}_r/4 + \boldsymbol{\beta})\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}\dot{\boldsymbol{\beta}}\boldsymbol{e}_p\Big/2 \\
&= \dot{\boldsymbol{e}}_p^{\mathrm{T}}(\boldsymbol{M}_r/2 - \boldsymbol{D}^*)\dot{\boldsymbol{e}}_p + \boldsymbol{e}_p^{\mathrm{T}}(\boldsymbol{M}_r/4 + \boldsymbol{\beta} - \boldsymbol{K}^* - \boldsymbol{D}^*/2)\dot{\boldsymbol{e}}_p \\
&\quad + \boldsymbol{e}_p^{\mathrm{T}}(2\dot{\boldsymbol{\beta}} - \boldsymbol{K}^*)\boldsymbol{e}_p\Big/2.
\end{aligned} \tag{36}$$

References

[1] Y. Yuan, H.-T. Zhang, Y. Wu, T. Zhu, and H. Ding, "Bayesian learning-based model-predictive vibration control for thin-walled workpiece machining processes," *IEEE/ASME Trans. Mechatron.*, vol. 22, no. 1, pp. 509–520, Feb. 2017.

[2] G. Wang, W.-I. Li, C. Jiang, and H. Ding, "Machining allowance calculation for robotic edge milling an aircraft skin considering the deformation of assembly process," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 5, pp. 3350–3361, Oct. 2022.

[3] W. J. Seong, Y. C. Jeon, and S. J. Na, "Ship-hull plate forming of saddle shape by geometrical approach," *J. Mater. Process. Technol.*, vol. 213, no. 11, pp. 1885–1893, Nov. 2013.

[4] L. Jiang, "Research on application of key technologies of error compensation for CNC machine tools based on computer," *J. Phys., Conf. Ser.*, vol. 1721, Oct. 2020, Art. no. 012035.

[5] A. Verl, A. Valente, S. Melkote, C. Brecher, E. Ozturk, and L. T. Tunc, "Robots in machining," *CIRP Ann.*, vol. 68, no. 2, pp. 799–822, Jun. 2019.

[6] Y. Dai, Y. Xue, and J. Zhang, "Milling state identification based on vibration sense of a robotic surgical system," *IEEE Trans. Ind. Electron.*, vol. 63, no. 10, pp. 6184–6193, Oct. 2016.

[7] X. Zhao, S. Han, B. Tao, Z. Yin, and H. Ding, "Model-based actor-critic learning of robotic impedance control in complex interactive environment," *IEEE Trans. Ind. Electron.*, vol. 69, no. 12, pp. 13225–13235, Dec. 2022.

[8] A. Klimchik, Y. Wu, S. Caro, B. Furet, and A. Pashkevich, "Accuracy improvement of robot-based milling using an enhanced manipulator model," *Adv. Theory Pract. Robots Manipulators Mechanisms Mach. Sci.*, vol. 22, pp. 73–81, Jan. 2014.

[9] L. Wang and H. Si, "Machining deformation prediction of thin-walled workpieces in five-axis flank milling," *Int. J. Adv. Manuf. Technol.*, vol. 97, no. 9, pp. 4179–4193, Jun. 2018.

[10] Z. Li, O. Tuysuz, L. Zhu, and Y. Altintas, "Surface form error prediction in five-axis flank milling of thin-walled parts," *Int. J. Mach. Tools Manufacture*, vol. 128, pp. 21–32, Feb. 2018.

[11] Q. Bi, X. Wang, Q. Wu, L. Zhu, and H. Ding, "Fv-SVM-based wall-thickness error decomposition for adaptive machining of large skin parts," *IEEE Trans. Ind. Inform.*, vol. 15, no. 4, pp. 2426–2434, Apr. 2019.

[12] N. Hogan, "Impedance control: An approach to manipulation. I: Theory," in *Proc. Amer. Control Conf.*, Jun. 1984, vol. 107, pp. 304–313.

[13] H. Ochoa and R. Cortesão, "Impedance control architecture for robotic-assisted mold polishing based on human demonstration," *IEEE Trans. Ind. Electron.*, vol. 69, no. 4, pp. 3822–3830, Apr. 2022.

[14] Y. Li, N. Jarrassé, and E. Burdet, "Versatile interaction control and haptic identification in humans and robots," *Geometric Numer. Foundations Movements*, vol. 117, pp. 187–206, 2017.

[15] R. Ikeura and H. Inooka, "Variable impedance control of a robot for cooperation with a human," in *Proc. IEEE Int. Conf. Robot. Automat.*, 1995, vol. 3, pp. 3097–3102.

[16] M. M. Rahman, R. Ikeura, and K. Mizutani, "Investigating the impedance characteristic of human arm for development of robots to co-operate with human operators," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 1999, vol. 2, pp. 676–681.

[17] S. Kozlovsky, E. Newman, and M. Zacksenhouse, "Reinforcement learning of impedance policies for peg-in-hole tasks: Role of asymmetric matrices," *IEEE Robot. Automat. Lett.*, vol. 7, no. 4, pp. 10898–10905, Oct. 2022.

[18] F. J. Abu-Dakka, L. Rozo, and D. G. Caldwell, "Force-based variable impedance learning for robotic manipulation," *Robot. Auton. Syst.*, vol. 109, 2018, Art. no. 156.

[19] R. Wu and A. Billard, "Learning from demonstration and interactive control of variable-impedance to cut soft tissues," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 5, pp. 2740–2751, Oct. 2022.

[20] Y. Li, G. Ganesh, N. Jarrass, S. Haddadin, A. Albu-Schaeffer, and E. Burdet, "Force impedance and trajectory learning for contact tooling and haptic identification," *IEEE Trans. Robot.*, vol. 34, no. 5, pp. 1170–1182, Oct. 2018.

[21] J. Buchli, F. Stulp, E. Theodorou, and S. Schaal, "Learning variable impedance control," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 820–833, 2011.

[22] X. Zhao, B. Tao, L. Qian, and H. Ding, "Model-based actor-critic learning for optimal tracking control of robots with input saturation," *IEEE Trans. Ind. Electron.*, vol. 68, no. 6, pp. 5046–5056, Jun. 2021.

[23] F. J. Abu-Dakka and M. Saveriano, "Variable impedance control and learning—A review," *Front. Robot. AI*, vol. 7, 2020, Art. no. 590681.

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[24] J. Colgate and H. Hogan, "Robust control of dynamically interacting systems," *Int. J. Control*, vol. 48, no. 1, pp. 65–88, 1988.
[25] F. Ferraguti et al., "An energy tank-based interactive control architecture for autonomous and teleoperated robotic surgery," *IEEE Trans. Robot.*, vol. 31, no. 5, pp. 1073–1088, Oct. 2015.
[26] K. Kronander and A. Billard, "Stability considerations for variable impedance control," *IEEE Trans. Robot.*, vol. 32, no. 5, pp. 1298–1305, Oct. 2016.
[27] Z. Jin, A. Liu, W.-A. Zhang, and L. Yu, "An optimal variable impedance control with consideration of the stability," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 1737–1744, Apr. 2022.
[28] E. S. Papastavridis and J. S. Dai, "Minimally model-based trajectory tracking and variable impedance control of flexible-joint robots," *IEEE Trans. Ind. Electron.*, vol. 68, no. 7, pp. 6031–6041, Jul. 2021.
[29] E. S. Papastavridis and J. S. Dai, "Stable flexible-joint floating-base robot balancing and locomotion via variable impedance control," *IEEE Trans. Ind. Electron.*, vol. 70, no. 3, pp. 2748–2758, Mar. 2023.
[30] E. S. Papastavridis, Z. Fu, and J. S. Dai, "Power-shaping model-based control with feedback deactivation for flexible-joint robot interaction," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 4566–4573, Apr. 2022.
[31] Y. Michel, C. Ott, and D. Lee, "Safety-aware hierarchical passivity-based variable compliance control for redundant manipulators," *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3899–3916, Dec. 2022.
[32] J. Duan, Y. Gan, M. Chen, and X. Dai, "Adaptive variable impedance control for dynamic contact force tracking in uncertain environment," *Robot. Auton. Syst.*, vol. 102, pp. 54–65, Apr. 2018.
[33] T. D. Barfoot, *State Estimation for Robotics*. Cambridge, U.K.: Cambridge Univ. Press, 2017.
[34] C. V. Rao, J. B. Rawlings, and D. Q. Mayne, "Constrained state estimation for nonlinear discrete-time systems: Stability and moving horizon approximations," *IEEE Trans. Autom. Control*, vol. 48, no. 2, pp. 246–258, Feb. 2003.
[35] O. Adiyatov, B. Rakhim, A. Zhakatayev, and H. A. Varol, "Sensor reduction of variable stiffness actuated robots using moving horizon estimation," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 5, pp. 1757–1769, Sep. 2020.
[36] P. Kühl, M. Diehl, T. Kraus, J. P. Schlöder, and H. G. Bock, "A real-time algorithm for moving horizon state and parameter estimation," *Comput. Chem. Eng.*, vol. 35, no. 1, pp. 71–83, Jan. 2011.
[37] T. Kraus et al., "Moving horizon estimation and nonlinear model predictive control for autonomous agricultural vehicles," *Comput. Electron. Agriculture*, vol. 98, pp. 25–33, Oct. 2013.
[38] A. Liu, W.-A. Zhang, M. Z. Q. Chen, and L. Yu, "Moving horizon estimation for mobile robots with multirate sampling," *IEEE Trans. Ind. Electron.*, vol. 64, no. 2, pp. 1457–1467, Feb. 2017.
[39] A. Alessandri, P. Bagnerini, M. Gaggero, D. Lengani, and D. Simoni, "Detection of flow-regime transitions using dynamic mode decomposition and moving horizon estimation," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 3, pp. 1324–1331, May 2021,.
[40] J. D. Schiller and M. A. Müller, "Suboptimal nonlinear moving horizon estimation," *IEEE Trans. Autom. Control*, vol. 68, no. 4, pp. 2199–2214, Apr. 2023.
[41] H. Seraji and R. Colbaugh, "Force tracking in impedance control," *Int. J. Robot. Res.*, vol. 16, no. 1, pp. 97–117, 1997.
[42] S. Jung, T. C. Hsia, and R. G. Bonitz, "Force tracking impedance control of robot manipulators under unknown environment," *IEEE Trans. Control Syst. Technol.*, vol. 12, no. 3, pp. 474–483, May 2004.
[43] D. Robertson, J. Lee, and J. Rawlings, "A moving horizon-based approach for least-squares estimation," *AIChE J.*, vol. 42, no. 8, pp. 2209–2224, 1996.
[44] R. Quirynen, M. Vukov, M. Zanon, and M. Diehl, "Autogenerating microsecond solvers for non-Standard nonlinear MPC: A tutorial using ACADO integrators," *Optim. Control Appl. Methods*, vol. 36, no. 5, pp. 685–704, Nov. 2014.
[45] J. P. Den Hartog, *Strength of Materials*. New York, NY, USA: Dover, 2012.
[46] M. S. Grewal and A. P. Andrews, *Kalman Filtering: Theory and Practice With MATLAB*. Hoboken, NJ, USA: Wiley, 2015.
[47] S. Gabl and F. Walch, "Franka control interface," Github.com. Accessed: Oct. 13, 2021. [Online]. Available: https://frankaemika.github.io/libfranka/motion_with_control_8cpp-example.html



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